

# System Documentation

Electronics

**GenoLine System with LOP 10**

**and MDEC**

**for Onboard Power Generators**

**Engine Series 2000 P**

Application: Marine

E532427/00E



*Power. Passion. Partnership.*

Printed in Germany

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This handbook is provided for use by maintenance and operating personnel in order to avoid malfunctions or damage during operation.

Subject to alterations and amendments.

# Commissioning Note

## Important

Please complete and return the “Commissioning Note” card below to MTU Friedrichshafen GmbH.  
The Commissioning Note information serves as a basis for the contractually agreed logistic support (warranty, spare parts, etc.).



**Postcard**

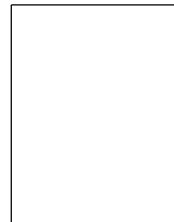
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**GERMANY**



|                                                                                                                                                          |                                      |
|----------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------|
| <b>Engine No.:</b>                                                                                                                                       | <b>MTU works order no.:</b>          |
| <b>Engine model:</b>                                                                                                                                     | <b>Date put into operation:</b>      |
| <b>Application: *</b><br><input type="checkbox"/> Marine <input type="checkbox"/> Rail<br><input type="checkbox"/> Genset <input type="checkbox"/> ..... | <b>Type:</b><br><b>Manufacturer:</b> |
| <b>End user's address:</b>                                                                                                                               |                                      |
| <b>Remarks:</b>                                                                                                                                          |                                      |

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# 1 General Provisions

## 1.1 Safety

### 1.1.1 General conditions

#### **a) General information**

In addition to the instructions in this publication, the applicable country-specific legislation and other compulsory regulations regarding accident prevention and environmental protection must be observed. This MTU engine is a state-of-the-art product and conforms with all the applicable specifications and regulations. Nevertheless, persons and property may be at risk in the event of:

- Incorrect use
- Operation, maintenance and repair by unqualified personnel
- Modifications or conversions
- Noncompliance with the Safety Instructions

#### **b) Intended use**

1. The engine is intended exclusively for the application specified in the contract or defined at the time of delivery. Any other use is considered improper use. The engine manufacturer will accept no liability for any resultant damage. The responsibility is borne by the user alone.
2. Intended use also includes observation of the Operating Instructions and compliance with the maintenance specifications.

#### **c) Modifications or conversions**

1. Unauthorized engine modifications compromise safety.
2. MTU will accept no liability or warranty claims for any damage caused by unauthorized modifications or conversions.

#### **d) Spare parts**

1. Only genuine MTU spare parts must be used to replace components or assemblies.
2. In event of any damage caused by the use of other spare parts, no liability or warranty claims vis-à-vis the engine manufacturer will be accepted.

## 1.1.2 Personnel and organizational requirements

### **Organizational measures of the operator**

This manual must be issued to all personnel involved in operation, maintenance, repair or transportation.

Keep this manual handy in the vicinity of the product such that it is accessible to operating, maintenance, repair and transport personnel at all times.

Use this manual as a basis for instructing personnel on product operation and repair, whereby the safety-relevant instructions, in particular, must be read and understood.

This is particularly important in the case of personnel who only occasionally perform work on or around the product.

- This personnel must be instructed repeatedly.

### **Personnel requirements**

All work on the product shall be carried out by trained and qualified personnel only:

- Training at the Training Center of the manufacturer
- Qualified personnel specialized in mechanical and plant engineering

The operator must define the responsibilities of the personnel involved in operation, maintenance, repair and transport.

### **Working clothes and personal protective equipment**

- Wear proper protective clothing for all work.
- Always wear appropriate personal protection when working. For example:
  - Ear protectors
  - Protective gloves
  - Industrial goggles
  - Breathing protection
  - Safety mask
- Observe the information on personal protective equipment in the respective activity description.

### 1.1.3 Fluids and lubricants, fire prevention and environmental protection

#### a) Fire prevention

1. Rectify any fuel or oil leaks immediately; even quantities of oil or fuel on hot components can cause fires – therefore always keep the engine in a clean condition. Do not leave rags soaked with fluids and lubricants on the engine. Do not store combustible materials near the engine.
2. Do not weld pipes and components carrying oil or fuel. Before welding, clean with a nonflammable fluid.
3. When starting the engine with an external power source, connect the ground cable last and remove it first. To avoid sparks in the vicinity of the battery, connect the ground cable from the external power source to the ground cable of the engine or to the ground terminal of the starter.
4. Always have a suitable extinguishant (fire extinguisher) on hand and familiarize yourself fully with its handling.

#### b) Noise

1. Noise can lead to an increased risk of accidents if acoustic signals, warning shouts or sounds indicating danger are drowned.
2. In all workplaces with a sound pressure level in excess of 85 dB (A), wear ear protectors (cotton wool, ear plugs or capsules).

#### c) Environmental protection

1. Dispose of used fluids, lubricants and filters in accordance with local regulations.
2. Manipulation of the injection or control system can influence the engine performance and exhaust emissions. As a result, compliance with environmental regulations may no longer be guaranteed.
3. Only fuels of the specified quality required to achieve emission limits must be used.

#### d) Fluids, lubricants and process materials

1. Use only fluids and lubricants that have been tested and approved by MTU.
2. Fluids, lubricants and process materials must be kept in suitable, correctly labeled containers.
3. When using fluids, lubricants, process materials and other chemical substances, follow the safety regulations that apply to the product concerned.
4. Take special care when using hot, chilled or caustic materials.
5. When using flammable materials, avoid all sparks and do not smoke.

#### e) Lead

1. When working with lead or leaded compounds, avoid direct contact to the skin and do not inhale lead vapors.
2. Prevent the buildup of white powder of lead.
3. Switch on fume extraction system.
4. After coming into contact with lead or lead-containing materials, wash your hands!

#### f) Acid and alkaline solutions

1. When working with acids and alkaline solutions, wear protective goggles or face mask, gloves and protective clothing.
2. If such solutions are spilled onto clothing, remove the affected clothing immediately!
3. Rinse injured parts of the body thoroughly with clean water!
4. Rinse eyes immediately with eyedrops or clean tap water!



### **g) Paints and lacquers**

1. When carrying out painting work outside the spray stands provided with fume extraction systems, ensure that the area is well ventilated. Make sure that neighboring work areas are not impaired.
2. No naked flames!
3. No smoking!
4. Observe all fire-prevention regulations!
5. Always wear a mask providing protection against paint and solvent vapors!

### **h) Liquid nitrogen**

1. Store liquid nitrogen only in small quantities and always in regulation containers (without fixed covers).
2. Avoid body contact (eyes, hands). Contact of this nature would cause frostbite and numbing.
3. Wear protective clothing, protective gloves, closed shoes and protective goggles / safety mask!
4. Make sure that working area is well ventilated. Suffocation will result at 88% contamination of breathing air with nitrogen.
5. Take great care not to subject containers, fittings and workpieces to impact or shock.

### **i) Compressed air**

1. Compressed air is air which has been compressed at excess pressure and is stored in tanks from which it can then be extracted.
2. The pressure at which the air is kept can be read off at pressure gages which must be connected to the compressed air tanks and the compressed air lines.
3. When working with compressed air, safety precautions must be constantly observed:
  - Pay special attention to the pressure level in the compressed air network and pressure vessel!
  - Connecting devices and equipment must either be assembled for this pressure, or, if the permitted pressure for the connecting elements is lower than the pressure required, a pressure reducing valve and safety valve (set to permitted pressure) must form an intermediate connection. Hose couplings and connections must be securely attached!
  - Always wear protective goggles when blowing off workpieces or extracting chips!
  - The snout of the air nozzle is provided with a protective disk (e.g. rubber disk), which prevents airborne particles being reflected and thereby prevents injury to eyes.
  - First shut off compressed air lines before compressed air equipment is disconnected from the supply line, or before equipment or tool is to be replaced!
  - Unauthorized use of compressed air, e.g. forcing flammable liquids (danger class A1, AII and B) out of containers, represents an explosion hazard!
  - Forcing compressed air into thin-walled containers (e.g. containers made of tin, plastic and glass) for drying purposes or to check for leaks, results in a risk of explosions!
  - Blowing dirt from soiled clothes while still worn is prohibited!

### **j) Used oil**

1. Used oil may contain harmful combustion residues.
2. Rub your hands with skin protection cream!
3. Wash your hands after contact with used oil.

#### 1.1.4 Standards for safety notices in the text

|                                                                                                     |                                                                                                                                                                                                                                         |
|-----------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>DANGER</b><br>  | In the event of immediate danger.<br><b>Consequences: Death, serious or permanent injury!</b> <ul style="list-style-type: none"><li>• Remedial action.</li></ul>                                                                        |
| <b>WARNING</b><br> | In the event of a situation involving potential danger.<br><b>Consequences: Death, serious or permanent injury!</b> <ul style="list-style-type: none"><li>• Remedial action.</li></ul>                                                  |
| <b>CAUTION</b><br> | In the event of a situation involving potential danger.<br><b>Consequences: Minor or moderate injuries!</b> <ul style="list-style-type: none"><li>• Remedial action.</li></ul>                                                          |
| <b>NOTICE</b><br>  | In the event of a situation involving potentially adverse effects on the product.<br><b>Consequences: Material damage!</b> <ul style="list-style-type: none"><li>• Remedial action.</li><li>• Additional product information.</li></ul> |

#### Safety notices

- This manual with all safety instructions and safety notices must be issued to all personnel involved in operation, maintenance, repair or transportation.